NOPROP

NoProp

Training Neural Networks without Full Back- or Forward-Propagation | CoLLAs 2025

Replace global credit assignment with block-local denoising: each block independently predicts a clean target representation from a noisy target and the input.

local learning · diffusion · parallelism

1 Problem

WHY LEAVE BACKPROP?

- Full backprop stores intermediate activations, propagates gradients sequentially and couples every block.
- This creates memory overhead, limits device-level parallelism and is biologically implausible.
- Many alternatives retain standard architectures but sacrifice accuracy or efficiency.

2 Core idea

LOCAL DENOISING

- Fix each block's target representation beforehand as a noised embedding of the class label.
- Broadcast input x to every block; train each block independently with only local backpropagation.
- No full forward or backward pass is needed during training, although inference still chains the blocks.

3 Training & inference

TWO DIFFERENT VIEWS

- Training samples a noise level and teaches block t to predict clean label embedding u_y from z_(t-1) and x.
- Inference starts at Gaussian noise z_0 and repeatedly moves the state toward each block's predicted label.
- Variants include discrete-time diffusion, continuous-time diffusion and flow matching.

4 Mathematics

LOCAL OBJECTIVE

- `z_t = a_t u_theta_t(z_(t-1),x) + b_t z_(t-1) + sqrt(c_t) epsilon_t`
- `L_local proportional to ||u_theta_t(z_(t-1),x) - u_y||^2`
- The full NoProp objective follows an ELBO; every dynamic block optimizes its own denoising term plus output classification.

5 Evidence

PROOF OF CONCEPT

- NoProp-DT is comparable to backprop: MNIST 99.54%, CIFAR-10 80.54%, CIFAR-100 46.06% with learned prototypes.
- CIFAR-10 training memory falls from 1.17 GB (backprop) to 0.64 GB; NoProp-CT uses 0.45 GB vs 6.23 GB for adjoint training.
- Continuous variants usually beat the Neural ODE adjoint baseline in accuracy and speed.

6 Meaning & limits

TAKEAWAY

- Credit assignment becomes modular, potentially distributed and more biologically plausible.
- Local gradient flow may help isolate task interference and inspire continual-learning methods.
- Evidence is limited to small classification datasets; broadcasting high-dimensional inputs may hinder scaling.